

Course: High Performance Computing Lab

Practical No 9

PRN: 22510056

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Batch: B6

Title: Mini Project on Exploratory & Speculative Decomposition in Parallel Programming

1. Introduction to Both Techniques

Exploratory Search

- Definition: Systematically explore all possible solutions without assuming any outcome.
- Use Case: When you don't know where the solution is — brute-force or guided search over entire space.
- Parallelization Strategy: Divide problem space (e.g., grid rows, columns, directions) among threads -> each explores a partition.
- No Redundancy: Each thread works on unique portion -> minimal wasted work.

Speculative Search

- Definition: Run multiple possible paths or strategies simultaneously — even if some become unnecessary — to gain speed.
- Use Case: When you want the answer *fast*, and can afford to discard redundant computation.
- Parallelization Strategy: Multiple threads search same space but with different strategies or starting points -> first to find wins.
- Wasted Work: Some threads keep searching after solution is found -> tracked as “discarded work”.

2. Problem descriptions

Part A — Exploratory Mini-Project:

Word Search Puzzle – Divide the grid among threads to search for words concurrently.

Part B — Speculative Mini-Project :

Speculative Search Strategy in 8-Puzzle

- a. For the sliding-tile puzzle, sequential search chooses either BFS or DFS.
- b. Speculative: run BFS and DFS simultaneously; stop both when the first valid solution is found.

Code 1:

```
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment9$ gcc part1.c -o part1 -fopenmp
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment9$ ./part1
=== Sequential ===
Time: 0.000278 s

=== Parallel ===
Time: 0.011057 s

=== Results ===
FINDING FOUND at (3,1) dir S
NEMO FOUND at (4,4) dir NE
ELLEN FOUND at (1,0) dir S
MARLIN FOUND at (1,9) dir SW
CLOWNFISH FOUND at (12,10) dir N
CORALREEF FOUND at (13,0) dir E
PELICAN FOUND at (0,3) dir SE
BARRACUDA FOUND at (12,0) dir E
DORY FOUND at (9,7) dir W
SHARKS FOUND at (11,0) dir N
BRUCE FOUND at (8,11) dir S
ANCHOR FOUND at (0,1) dir SE
CHUM FOUND at (9,9) dir NW
DENTIST FOUND at (10,3) dir E
TANKGANG FOUND at (10,9) dir NW
GILL FOUND at (5,7) dir NE
BLOAT FOUND at (8,11) dir N
BUBBLES FOUND at (11,3) dir E
PEACH FOUND at (0,6) dir SW
GURGLE FOUND at (6,2) dir S
DEB FOUND at (8,3) dir E
soham@Vivobook16x-Soham:/mnt/d/SEM VII/HPCL/Assignment9$
```

Github link : [🌐 HPCL/Assignment9 at main · Somzee5/HPCL](https://github.com/Somzee5/HPCL/tree/main/Assignment9)

Class: Final Year (CSE)

Year: 2025-26 Semester: 1